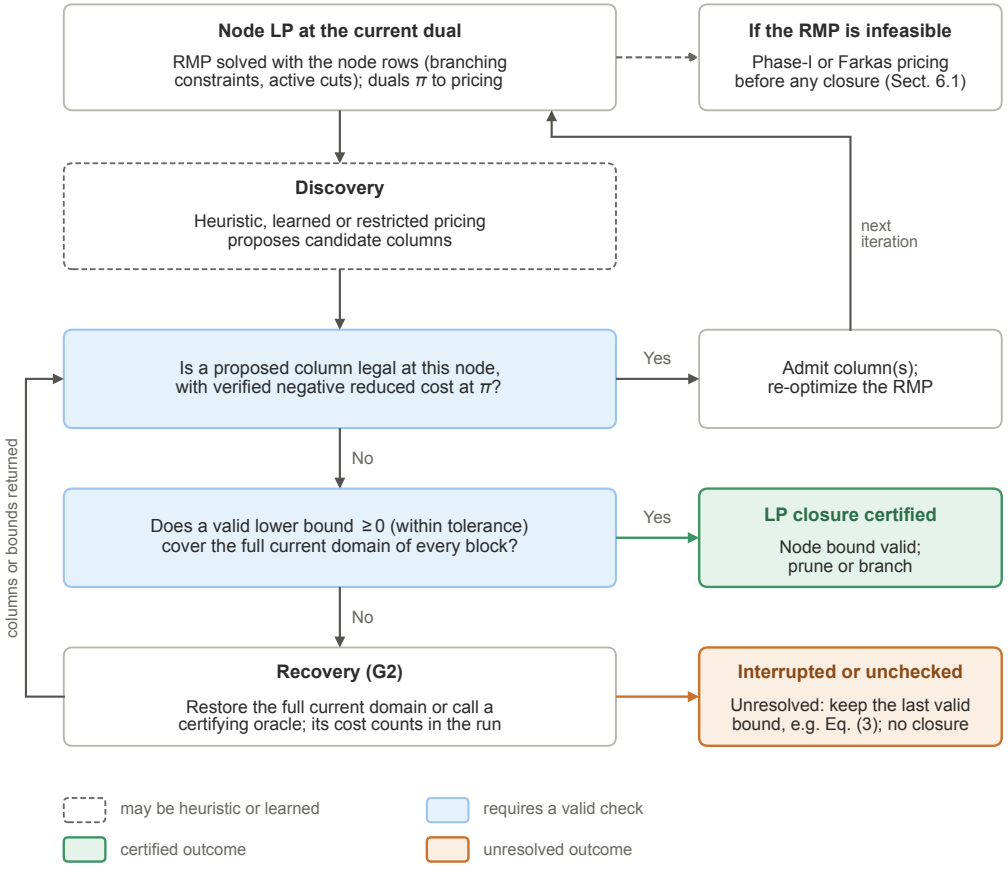
Node LP at the current dual
RMP solved with the node rows (branching constraints, active cuts); duals π to pricing
If the RMP is infeasible
Phase-I or Farkas pricing before any closure (Sect. 6.1)
Discovery
Heuristic, learned or restricted pricing proposes candidate columns
next iteration
Is a proposed column legal at this node, with verified negative reduced cost at π?
Yes
Admit column(s); re-optimize the RMP
No
columns or bounds returned
Does a valid lower bound ≥ 0 (within tolerance) cover the full current domain of every block?
Yes
LP closure certified
Node bound valid; prune or branch
No
Recovery (G2)
Restore the full current domain or call a certifying oracle; its cost counts in the run
Interrupted or unchecked
Unresolved: keep the last valid bound, e.g. Eq. (3); no closure
may be heuristic or learned
requires a valid check
certified outcome
unresolved outcome